

Track A: Biosynthesis Pathway

Question: Which enzymes synthesize metabolites?

Method: GNN (HGT) → Proteomics Validation

Result: PAL→4CL→CHS→CHI→IFS→IFR all ↑

✓ VALIDATED



Track B: Metabolite-Protein Binding

Question: Do metabolites regulate proteins?

Method: Molecular Docking

Result: Daidzein-FNR, Formononetin-Kinase

⚠ HYPOTHETICAL

KEY MESSAGE

These tracks address DIFFERENT biological questions:

- Track A: Pathway membership (enzyme synthesizes metabolite)
- Track B: Physical binding (metabolite binds protein)

Docking CANNOT validate GNN predictions!

(Enzymes bind substrates, not products)